Synchronized arithmetic certificates for ZFC refutability

An explicit quartic in 522 nonnegative integers,
and a quartic in 2,088 unrestricted integers

Prepared for Vladimir Reshetnikov's ProveIt project
AI-assisted mathematical research and reproducible algebra

24 October 2026

Abstract

We give a fully materialized polynomial $P_{522} \in \mathbb{Z}[x_0, \ldots, x_{521}]$ of exact degree four such that

$$\mathrm{ZFC} \vdash \left(\nexists x \in \mathbb{N}^{522} : P_{522}(x) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}), \qquad \mathbb{N} = \{0, 1, 2, \ldots\}.$$

It is the sum of squares of 400 explicitly listed quadratic residuals, with 20,225 nonzero expanded monomials and coefficient height 83,900,628. There are no unbound coefficient ports, variable exponents, factorial macros, or bounded quantifiers in the delivered polynomial. An accompanying polynomial $P^{\mathbb{Z}}_{2088}$ has 2,088 unrestricted integer variables and exact degree four, with the same consistency equivalence.

The construction directly encodes the calibrated numerical calculus C_{222}; it does not simulate K215 or invoke Jones's universal equation. Its main new step is to synchronize arithmetic certificates across an entire finite circuit: one Pell frame handles every exponentiation, one auxiliary Pell solution handles every index, and one delayed radix handles every binomial coefficient. A circuit majorant replaces a large duplicate evaluation, and certified polynomial substitutions remove unnecessary coordinates. Against the newest inspected direct-certificate archive, the natural-variable count falls from 3,614 to 522, a reduction of 85.56%, while the maximum coefficient also decreases.

These are ordinary mathematical proofs formalizable in ZFC, accompanied by exact finite algebra audits. We do not claim global minimality, worldwide priority, a proof of consistency, or a newly compiled end-to-end Lean or Rocq certificate.

1 Result, comparison, and exact deliverable

The comparison snapshot is ProveIt commit `a88aefd7ddab4771534bb049c67350e397c91b9a`. Its active research notes and incoming archives contain stronger results than the numbered collection's opening summary. In particular, the incoming `ConZFC_Shared_Certificates_2026-10-24.zip` already supplies a direct quartic in 3,614 natural variables. That archive, rather than the older 16,036-variable direct construction, is our immediate baseline [1, 2, 3].

Theorem 1.1 (Delivered equations)**.** *Let P_{522} be the literal expanded polynomial in* `equations/natural522/expanded_polynomial.json`*, with the coordinate ordering in the accompanying* `equation.json`*. Then*

$$\mathrm{ZFC} \vdash \Big(\exists x \in \mathbb{N}^{522}\ P_{522}(x) = 0\Big) \longleftrightarrow (\exists p\ \mathrm{Prf}_{\mathrm{ZFC}}(p, \bot))\,. \tag{1}$$

The polynomial has exact degree four, 20,225 monomials, and coefficient height 83,900,628. Its compact, literal definition is

$$\boxed{P_{522}(x) = \sum_{j=0}^{399} R_j(x)^2 = 0, \qquad x \in \mathbb{N}^{522},} \tag{2}$$

where every R_j is explicitly supplied and has degree at most two. The 529- and 596-coordinate companion quartics have the same equivalence. The polynomial $P^{\mathbb{Z}}_{2088}$ in `equations/integer2088` *satisfies* (1) *with $x \in \mathbb{Z}^{2088}$ in place of $x \in \mathbb{N}^{522}$. Taking negations gives the requested no-solution/consistency equivalence.*

The proof occupies the following sections. The numerical measurements are obtained by exact expansion and coefficient collection; they are not estimates from the syntax of an arithmetic circuit.

Natural quartic	Variables	Residuals	Monomials	Coefficient height
Incoming direct baseline	3,614	2,954	15,996	127,778,545
Synchronized source	895	773	4,863	63,908,265
Sparse contraction	596	474	7,263	63,909,160
Compact contraction	529	407	10,782	211,184,624
Fewest variables delivered	**522**	**400**	20,225	83,900,628

Table 1: All rows have exact degree four and quantify nonnegative integers. The different support limits are deliberate expression-size tradeoffs.

The 522-coordinate form minimizes variables among the variants materialized here. The 529-coordinate form uses only seven more coordinates and about half as many monomials. The 596-coordinate form has a smaller coefficient height and fewer monomials still. These are different useful choices; no optimization theorem for the contraction order is asserted.

There are already much lower-variable equations in the repository. Its active comparison records nine-positive and ten-integer equations, with much larger degree, and a separate incoming norm construction has thirteen integer variables. Older machine-derived quartics also have far fewer variables, at the cost of large fixed coefficient payloads. Our result improves the *direct, fully expanded, modest-coefficient quartic route*. It is not a replacement for every point of that broader variable/degree/coefficient frontier. Jones's and Sun's reductions remain relevant background, but neither is used in this construction [2, 9, 10].

1.1 What makes the polynomial explicit

Each residual file contains the complete ordered variable list and a finite list of integer coefficients and monomials. A term is encoded as `[coefficient,[variable_indices]]`; an empty index list denotes the constant monomial and repeated indices denote powers. Indices start at zero. The expanded file is the same polynomial after exact squaring, summation, and collection. All coefficients are ordinary signed decimal integers; for P_{522}, their magnitudes are less than 2^{27}.

The archive includes a deterministic generator, independently implemented checks, the full elimination maps, immediate source dependencies, and cryptographic file ledgers. The hashes identify the supplied bytes; the literal data and mathematical construction, not a hash alone, define the equation. The construction never runs a consistency-deciding search and never uses an answer to the consistency question in choosing a coefficient.

2 Logical source: a numerical calculus rather than a machine

The inherited source is the numerical calculus C_{222}. Its false target is code 1, and its freely available truth label is code 0. The repository supplies unconditional calibration statements

$$C_{222}(1) \longleftrightarrow \ \mathrm{BProv}(\mathrm{ZF}, \bot) \ \longleftrightarrow \ \mathrm{BProv}(\mathrm{ZFC}, \bot). \tag{3}$$

The named sources are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`; the latter uses the constructible-universe argument for Choice. Their immediate source files and pinned hashes are included [4, 5]. The first theorem is independent of the machine parameters that occur later in its source module. Some imported modules also support machine theorems; we make no claim that the entire import graph is machine-free.

For completeness, here is the entire arithmetic interface being encoded. Write

$$\begin{aligned} I(a,b) &= 2(a+b)(a+b+1)+4b, \\ \mathsf{M}(a,b) &= I(a,b)+1, & \mathsf{A}(a,b) &= I(a,b)+2, & \mathsf{E}(a,b) &= I(a,b)+3, \\ \mathsf{N}(a) &= I(a,1), & \mathsf{X}(a,b) &= \mathsf{N}(\mathsf{A}(a,\mathsf{N}(b))). \end{aligned}$$

The eight unconditional axiom codes are

$$\begin{aligned} c_1 &= \mathsf{X}(a,\mathsf{E}(a,b)), \\ c_2 &= I(\mathsf{A}(a,I(b,c)),I(\mathsf{A}(a,b),\mathsf{A}(a,c))), \\ c_3 &= I(\mathsf{E}(a,b),I(\mathsf{M}(c,a),\mathsf{M}(c,b))), \\ c_4 &= I(\mathsf{A}(a,\mathsf{A}(b,c)),\mathsf{A}(b,\mathsf{A}(a,c))), \\ c_5 &= I(\mathsf{E}(c,a),I(\mathsf{M}(c,b),\mathsf{M}(a,b))), \\ c_6 &= I(\mathsf{X}(a,\mathsf{A}(a,b)),b), \\ c_7 &= I(\mathsf{E}(c,a),I(\mathsf{E}(c,b),\mathsf{A}(\mathsf{M}(a,b),\mathsf{E}(a,b)))), \\ c_8 &= I(I(I(c,d),a),I(I(a,c),I(b,c))). \end{aligned}$$

The ninth, guarded bundle code is a fixed composition. To avoid an unreadable nested expression, define the following polynomial abbreviations:

$$\begin{aligned} \mathsf{R}(b,k,h) &= \mathsf{A}(b,I(h,\mathsf{N}(\mathsf{A}(b,I(h,\mathsf{N}(\mathsf{M}(b,k))))))), \\ \mathsf{S}(a,b,k) &= \mathsf{A}(k,I(\mathsf{M}(k,a),\mathsf{M}(k,b))), \\ C_0(a,b,k,h) &= I(\mathsf{S}(a,b,k),\mathsf{R}(b,k,h)), \\ C_1(a,b,k,h) &= \mathsf{N}(\mathsf{A}(a,C_0(a,b,k,h))), \\ C_2(a,b,k,h) &= \mathsf{N}(\mathsf{A}(a,I(h,C_1(a,b,k,h)))), \\ \mathsf{C}(a,b,k,h) &= \mathsf{A}(b,I(\mathsf{M}(b,k),C_2(a,b,k,h))), \\ \mathsf{V}(a,b,c) &= I(\mathsf{M}(a,b),\mathsf{A}(c,\mathsf{M}(a,b))), \\ \mathsf{K}(a,b,c) &= \mathsf{A}(c,I(\mathsf{C}(a,b,c,c),\mathsf{N}(\mathsf{R}(b,c,c)))), \\ c_* &= \mathsf{N}(I(\mathsf{V}(a,b,c),\mathsf{K}(a,b,c))). \end{aligned}$$

Its side condition is represented by the single equation

$$g = (c-a)^2(c-b)^2-(d+1)^2 = 0.$$

Over $\mathbb{N}$, existence of d is equivalent to $c \neq a$ and $c \neq b$: take $d = |(c-a)(c-b)|-1$ in the converse direction.

There are twelve padded two-premise branches: free truth; the eight unconditional axioms; the guarded bundle; modus ponens $q = I(p, v)$; and generalization $q = 0$, $v = \mathsf{A}(a, p)$. In every axiom branch $p = q = 0$. The four schema coordinates a, b, c, d need satisfy only the selected branch. Exact compositional checks match all 56 inherited wire definitions and all 34 inherited branch residuals against these formulas.

2.1 Factoring the common axiom side condition

Put

$$G = p^2 + q^2, \qquad L = v \prod_{j=1}^{8} (v - c_j), \qquad H_* = (v - c_*)^2 + g^2.$$

The union of the ten axiom/truth branches is represented by

$$G + L^2 H_* = 0. \tag{4}$$

The quantities G, L^2, and H_* are nonnegative, so this equation forces $p = q = 0$ and either $L = 0$ or $H_* = 0$. These are exactly the required unconditional or guarded alternatives. Multiplying by the unchanged modus-ponens and generalization factors gives the full Horn-rule polynomial

$$\mathcal{H} = (G + L^2 H_*)\,(q - I(p, v))^2 \big(q^2 + (v - \mathsf{A}(a, p))^2\big). \tag{5}$$

This changes polynomial values away from zero. The claimed preservation is an exact equivalence of zero sets, not a polynomial identity.

3 Finite proofs as bounded arithmetic paths

We recall the inherited commutative-flow argument because its logical role matters. For a two-premise rule $(p, q) \to v$, assign the vector $e_v - e_p - e_q$ in the free abelian group on nonzero labels; put $e_0 = 0$. A finite proof tree of target 1 yields a multiset of rule instances whose total flow is e_1.

Conversely, let D be the least set containing truth and closed under the rules. If $1 \notin D$, sum the coordinates outside D. A rule whose conclusion is outside D has at least one premise outside D, so its contribution is nonpositive. A rule with conclusion in D also has nonpositive contribution. A nonnegative sum of rule vectors therefore cannot equal e_1, whose outside-D sum is one. Only the finitely many labels in the certificate are involved. Thus the flow certificate is equivalent to derivability, even though its rule multiset is not ordered.

3.1 A polynomial weight carrier

Let $t \geq 1$ be the number of rule occurrences, and choose $m \geq 1$ to bound all seven local Horn coordinates p, q, v, a, b, c, d. Set

$$M = (m + 1)^t, \qquad w(a) = 1 + aM.$$

For t rule instances define

$$A_0(X) = \prod_{i<t} (1 + v_i X), \qquad C_0(X) = (1 + X) \prod_{i<t} (1 + p_i X)(1 + q_i X).$$

If $A_0(M) = C_0(M)$, then $\deg C_0 \leq t$: otherwise $C_0(M) \geq M^{t+1}$ whereas $(1+mM)^t < ((m+1)M)^t = M^{t+1}$. Both polynomials now have coefficient sum at most $(m + 1)^t = M$, and constant coefficient

one, so every coefficient is strictly smaller than M. Uniqueness of base-M expansion gives $A_0 = C_0$. Unique factorization in $\mathbb{Q}[X]$ gives precisely the required flow vector. Numerical coprimality of the weights is not needed.

It suffices to encode an accumulator path

$$S_0 = M + 1, \quad S_t = 1, \qquad S_{i+1}(1 + v_i M) = S_i(1 + p_i M)(1 + q_i M). \tag{6}$$

All states are positive by induction. Multiplication and cancellation yield $A_0(M) = C_0(M)$. Conversely, expand a finite derivation into a tree and traverse it by replacing a frontier conclusion by its premises. The product of frontier weights supplies the required natural states.

3.2 Eleven coordinates per local row

Encode the path by the usual beta digits

$$\beta(x, y, i) = x \bmod D_i, \qquad D_i = 1 + (i + 1)y.$$

For a row $i < t$, the eleven local coordinates and their bounds are

Role	Coordinates	Bound
Horn rule and schemas	p, q, v, a, b, c, d	m
Consecutive states	S, S'	y
Beta quotients	u, u'	x

The three local equalities, besides $\mathcal{H} = 0$, are

$$S'(1 + vM) = S(1 + pM)(1 + qM), \tag{7}$$

$$x = S + D_i u, \qquad x = S' + (D_i + y)u'. \tag{8}$$

Because $S, S' \le y < D_i, D_i + y$, these are exact remainder equations. The two former remainder-slack coordinates are unnecessary. The quotient bounds are valid since $x = S + D_i u$ with $D_i \ge 1$ implies $u \le x$, and similarly for u'.

Write $M = M_0 + 1$. The outer endpoint equations are

$$x = M_0 + 2 + (y + 1)u_0, \qquad y = M_0 + 2 + r_0, \tag{9}$$

$$x = 1 + (1 + (t + 1)y)u_t. \tag{10}$$

Thus $y \ge M + 1 \ge 2$, so the initial state is below its modulus and the final state 1 is below its modulus. An additional final remainder slack is unnecessary.

Completeness has an essential order of choices. First take the finite Horn proof and choose m to bound all seven proof coordinates. Then choose M and construct its finite state sequence. Choose y above all states and divisible by $t!$. The moduli $D_0, \ldots, D_t$ are pairwise coprime: a common divisor for D_i, D_j divides $j - i$ and is coprime to y, whereas $j - i \mid t! \mid y$. CRT supplies a natural x with every required state as its digit; taking a sufficiently large representative gives natural quotients. There is no independent uniform local bound and no circular requirement that y exceed its own code x.

Let P_{loc} be (5) plus the squares of the three residuals in (7)–(8). Then it is nonnegative, and zero precisely for a valid local row. Its large degree is an intermediate property; we subsequently represent its finite arithmetic circuit by quadratic gate equations.

4 Bounded universal checking and a cheap majorant

A signed polynomial circuit can be split as $P = E - F$, with E, F having nonnegative coefficients. Recursively use

$$(E_1, F_1)(E_2, F_2) = (E_1E_2 + F_1F_2,\ E_1F_2 + F_1E_2),$$

with componentwise addition and swapped components for negation. Identical subexpressions are shared. Apply this to P_{loc}. Let $z = (t, x, y, m, M_0)$, and let $\mathbf{u}$ be the eleven bounds above.

Lemma 4.1 (Shared bounded-row certificate)**.** *Suppose H is a natural integer satisfying*

$$H \geq E(t, z, \mathbf{u}) + F(t, z, \mathbf{u}) + t + x + y + m + 1.$$

Put $T = H!$ and $\mathcal{B} = \prod_{i<t}(1 + (i+1)T)$. Then

$$\forall i < t\ \exists w \leq \mathbf{u}:\ E(i, z, w) = F(i, z, w)$$

is equivalent to existence of natural $C, A_1, \ldots, A_{11}$ satisfying

$$\mathcal{B} \mid \binom{A_j}{u_j + 1} \qquad (1 \leq j \leq 11), \tag{11}$$

$$\mathcal{B} \mid 1 + (C+1)T, \tag{12}$$

$$E(C, z, A) \equiv F(C, z, A) \pmod{\mathcal{B}}. \tag{13}$$

All A_j may be required strictly positive.

Proof. Let $d_i = 1 + (i+1)T$. The d_i are pairwise coprime by the same gcd argument as for beta digits, and each is coprime to T. Every prime $\ell \mid d_i$ exceeds H, since a prime at most H divides $H!$. The identity

$$(u_j + 1)!\binom{A_j}{u_j + 1} = \prod_{r=0}^{u_j}(A_j - r)$$

shows from (11) that A_j is congruent modulo ℓ to some $w_j \in \{0, \ldots, u_j\}$. From (12), cancellation of T gives $C \equiv i \pmod{\ell}$. Equation (13) then gives equality of $E(i, z, w)$ and $F(i, z, w)$ modulo ℓ. Both are nonnegative and smaller than ℓ, by monotonicity and the bound on H, hence they are equal as integers.

Conversely, choose a witness for each of the finitely many rows and apply CRT modulo the pairwise coprime d_i to encode its coordinates and the index. A vanishing falling-factorial factor gives $\mathcal{B} \mid (u_j + 1)!\binom{A_j}{u_j+1}$. Since $(u_j + 1)! \mid H!$ and $\gcd(\mathcal{B}, H!) = 1$, cancel the factorial. The other two congruences follow directly. Adding positive multiples of $\mathcal{B}$ makes all coordinate representatives positive without affecting any of these arguments. □

4.1 Replacing a duplicate circuit by 23 squares

The baseline evaluates a large copy of $E + F$ at the bounding inputs to construct H. We need only an upper bound. Put

$$R_0 = 2 + t + x + y + m + M_0.$$

Every input is at most R_0, and $R_0 \geq 2$. Give each input exponent one and each constant $c \geq 0$ an exponent $e(c)$ with $c \leq 2^{e(c)}$. Use $e(0) = e(1) = 0$, and otherwise $e(c) = \lceil \log_2 c \rceil$. At a sum gate use $\max(e_1, e_2) + 1$; at a product gate use $e_1 + e_2$. Induction proves that every node is at most R_0 to

its assigned exponent. This induction applies to the actual nonnegative pair circuit, not to a signed polynomial after cancellation.

The supplied majorant certificate checks 388 gate inequalities for $E+F+t+x+y+m+1$ and gives exponent $6,893,159$. Therefore the convenient choice

$$\boxed{H = R_0^{8,388,608} = R_0^{2^{23}}} \tag{14}$$

is sufficient and uses only 23 squarings. This is a fixed polynomial power, subsequently represented by 23 quadratic defining equations. It is not a variable exponent left inside the final equation.

5 One radix for every binomial coefficient

The bounded certificate still contains binomial coefficients. The direct compiler has fourteen such requests, including those used to represent the two factorials and the modulus product. Treat their inputs and outputs as a finite semantic arithmetic circuit.

Theorem 5.1 (Simultaneous binomial digit certificate)**.** *For a finite family of positive upper arguments n_j and positive lower arguments k_j, set*

$$N = \sum_{n \in \{n_j\} \text{ as distinct circuit atoms}} n, \qquad q = 2^N + 1.$$

For each request introduce $C_j, L_j, U_j, h_j, r_j, s_j, d_j \in \mathbb{N}$ and impose

$$L_j = q^{k_j}, \qquad U_j = (q+1)^{n_j}, \tag{15}$$

$$U_j = L_j(qh_j + C_j) + r_j, \tag{16}$$

$$L_j = r_j + s_j + 1, \qquad q = C_j + d_j + 1. \tag{17}$$

Then these equations are equivalent to $C_j = \binom{n_j}{k_j}$ for every j. Equal power requests are represented once.

Proof. Each $n_j \le N$, so every binomial coefficient of $(1+X)^{n_j}$ is at most $2^{n_j} < q$. Hence $(q+1)^{n_j}$ has a carry-free base-q expansion with exactly those coefficients as digits. Equations (16)–(17) isolate digit k_j. This also handles $k_j > n_j$, when the digit is zero. Conversely take the ordinary quotient, remainder, and selected digit of that expansion; all slack variables are natural. □

The crucial implementation choice is to *defer* the digit certificates. A request for $\binom{n}{k}$ first allocates its output coordinate. After all requests are known, the generator constructs one common q and the digit equations. An upper argument can depend on another binomial's semantic output; this does not invalidate the theorem, which holds simultaneously for arbitrary natural assignments to all its arguments. The auxiliary radix is allowed to be chosen after the semantic values.

In the emitted system, fourteen binomial requests require one common radix, shared denominators, and shared power-index certificates. The total number of distinct power requests in the entire compiler drops from 55 in the baseline to 27. The next section certifies these 27 requests together.

6 One Pell frame for an entire power circuit

For $a \ge 2$, define the Pell sequences by

$$X_0(a) = 1, \qquad Y_0(a) = 0,$$
$$X_{j+1}(a) = aX_j(a) + (a^2-1)Y_j(a), \qquad Y_{j+1}(a) = X_j(a) + aY_j(a).$$

They satisfy $X_j(a)^2 - (a^2-1)Y_j(a)^2 = 1$. We use the standard classification of natural solutions of this norm equation and the finite Matiyasevich Pell-index criterion. These statements are available in the cited Mathlib development and its repository interface [6, 7, 8]. The full inherited exponentiation criterion is not needed: the shared frame supplies its own exponential bound.

6.1 The common exponential bound

For every distinct circuit atom occurring as a base n, exponent k, or proposed output m of a power request, define

$$W = 2 + \sum \text{these atoms}, \qquad Z = z + 1, \qquad A^2 - W^3(W+2)Z^2 = 1. \tag{18}$$

All coordinates in this section are natural. For each distinct base n introduce a natural modulus T_n with

$$2An = T_n + n^2 + 1. \tag{19}$$

Equation (19) excludes $n = 0$ in any solution. The positivity of every exponent k is a separate, checked syntactic property of the source. Thus $1 \le n, k \le W$ and $0 \le m \le W$.

Lemma 6.1 (The frame is exponentially large)**.** *Every solution of* (18) *and* (19) *has*

$$A \ge (W+1)^W > W^W, \qquad T_n \ge 2A - 2 > W^W \ge n^k.$$

In particular, $m < T_n$ for every power request. Conversely, after any finite collection of intended power values is fixed, a common frame exists.

Proof. The norm in (18) is $A^2 - ((W+1)^2 - 1)(WZ)^2 = 1$. Pell classification gives $A = X_j(W+1)$ and $WZ = Y_j(W+1)$. Reducing the recurrences modulo W gives $X_j(W+1) \equiv 1$ and $Y_j(W+1) \equiv j$. Hence $W \mid j$. Since $WZ > 0$, we have $j \ge W$. The recurrence gives $X_j(W+1) \ge (W+1)^j$, proving the exponential bound.

Also $A > W^2$ follows directly from the norm. The expression $2An - n^2 - 1$ is increasing for $1 \le n \le W$, and its value at one is $2A - 2$. Since $W \ge 2$, this is larger than W^W, as required. For completeness choose

$$A = X_W(W+1), \qquad Z = Y_W(W+1)/W.$$

The congruence above makes Z a positive integer and supplies the norm. □

The estimate $A > W^2$ by itself would not justify power recovery. The divisibility of the Pell index by W, and hence the exponential estimate in Lemma 6.1, is essential.

6.2 One auxiliary Pell solution for every index

There are 22 distinct exponent atoms among the 27 requests. For each one, denoted k_i, introduce X_i, Y_i, S_i, R_i and natural slacks e_i, s_i, r_i. Use one common auxiliary triple U, V, B and impose

$$Y_i = k_i + e_i, \qquad X_i^2 - (A^2-1)Y_i^2 = 1, \tag{20}$$

$$S_i^2 - (B^2-1)R_i^2 = 1, \qquad S_i = X_i + Us_i, \qquad R_i = k_i + 4Y_i r_i. \tag{21}$$

The common constraints are

$$U^2 - (A^2-1)V^2 = 1, \tag{22}$$

$$V = \left(\prod_{i=1}^{22} Y_i^2\right)(v_0 + 1), \tag{23}$$

$$B = 1 + 4V(b_0+1), \qquad B = A + Ub_1. \tag{24}$$

The product in (23) has the fixed length 22; it becomes a fixed list of quadratic multiplication gates in the final polynomial.

Lemma 6.2 (Simultaneous index correctness)**.** *Constraints* (20)–(24) *force* $X_i = X_{k_i}(A)$ *and* $Y_i = Y_{k_i}(A)$ *for every* i*. Conversely, for any finite family of positive* k_i *and* $A \geq 2$*, these constraints have simultaneous natural witnesses with those Pell coordinates.*

Proof. For soundness, $Y_i \geq k_i \geq 1$, $V > 0$, and $Y_i^2 \mid V$. Moreover $B > 1$, $B \equiv 1 \pmod{4Y_i}$, and $B \equiv A \pmod{U}$. Together with (20)–(21), these are exactly the nonzero branch of the finite Matiyasevich index criterion.

For completeness, first set $X_i = X_{k_i}(A)$ and $Y_i = Y_{k_i}(A)$. Let $L = 4\prod_i Y_i^2$. In the finite ring

$$(\mathbb{Z}/L\mathbb{Z})[s]/(s^2 - (A^2 - 1)),$$

the element $A + s$ is a unit, with inverse $A - s$. A positive power is therefore one. For its positive exponent r, put $U = X_r(A)$, $V = Y_r(A)$. Then $U \equiv 1 \pmod{L}$, $V \equiv 0 \pmod{L}$, and $V > 0$. In particular U is odd, and the Pell norm implies $\gcd(U, V) = 1$. Thus $\gcd(U, 4V) = 1$.

CRT gives $B \equiv A \pmod{U}$ and $B \equiv 1 \pmod{4V}$. Take a sufficiently large representative $B > A$. This supplies the nonnegative quotients in (24) and (23). Set $S_i = X_{k_i}(B)$ and $R_i = Y_{k_i}(B)$. The Pell recurrences are polynomial in the base, so $S_i \equiv X_i \pmod{U}$. Also $Y_k(1) = k$, so $R_i \equiv k_i \pmod{4Y_i}$. Monotonicity in the base gives $S_i \geq X_i$, and $Y_k(B) \geq k$ gives $R_i \geq k_i$. All displayed quotients and slacks are therefore natural. □

6.3 Recovering each power with a nonnegative quotient

For a request with exponent k_i, base n, and output m, impose one additional equation, with $h \in \mathbb{N}$:

$$X_i + nY_i = AY_i + m + T_n h. \tag{25}$$

For the true Pell sequences put $H_j = X_j(A) - (A - n)Y_j(A)$. Direct substitution gives

$$H_0 = 1, \qquad H_{j+1} = nH_j + T_n Y_j(A).$$

Consequently the exact identity

$$H_k - n^k = T_n \sum_{j=0}^{k-1} n^{k-1-j} Y_j(A) \tag{26}$$

has a nonnegative quotient. Soundness of (25) now follows from Lemmas 6.1 and 6.2: m and n^k are congruent modulo T_n and both lie in $[0, T_n)$, so they are equal. Completeness uses the sum in (26) as h.

The compiler shares the index block when the exponent atom is identical, the modulus when the base atom is identical, and the complete request when both are identical. It does not identify distinct expressions merely because they happen to have the same value in a particular assignment.

6.4 Why global sharing is not circular

For soundness, (19) first forces every base positive. All exponent arguments are positive by arithmetic syntax: $t = t_0 + 1$, $H = R_0^{2^{23}}$, shifted factorial arguments, positive encoded CRT coordinates, lower bounds plus one, and the sum of positive upper arguments. The entire power network is therefore correct as a relation on arbitrary proposed values. Next Theorem 5.1 identifies all binomial outputs

simultaneously. Only then is induction on the original semantic macro circuit used to identify factorials and the modulus product. No factorial positivity is assumed to establish the power network: even a factorial output proposed as a zero base is excluded by (19) itself.

For completeness, evaluate the original finite semantic circuit on its true inputs. Choose the common radix from those upper arguments and evaluate its associated powers. All 27 power requests now have true, finite values. Choose W and its Pell frame; then choose the common index certificate. Auxiliary scale choices occur after the semantic values they bound. Their size is unrestricted, as is appropriate for an existential Diophantine certificate.

7 Eliminating factorials, products, and arithmetic gates

We retain the elementary arithmetic graph identities of the incoming direct construction, with their power and binomial requests now handled by the shared machinery.

For $n \geq 1$, put

$$\rho = (n+2)^{n+2}, \qquad c = \binom{\rho+n}{n}, \qquad P = (\rho+1)^n.$$

Then $n! = \lceil P/c \rceil$. For $n \geq 2$,

$$\frac{P}{c} = n! \prod_{i=1}^{n} \frac{\rho+1}{\rho+i}, \qquad 0 \leq n! - \frac{P}{c} < \frac{n!n(n-1)}{2(\rho+1)} < 1;$$

the case $n = 1$ is immediate. Hence a proposed $f = n!$ is characterized by two natural remainder slacks and

$$P + c = cf + r + 1, \qquad c = r + s + 1.$$

These encode $c(f-1) < P \leq cf$.

The modulus product in Lemma 4.1 is also eliminated. Let

$$Q = (1 + (t+1)T)^t + 2, \qquad L = TQ - 1.$$

The exact identity is

$$\mathcal{B} = \left(T^t \, t! \binom{Q+t}{t} \right) \bmod L. \tag{27}$$

The numerator is $\prod_{i=1}^{t}(TQ + iT)$ and reduces modulo L to $\prod_{i=1}^{t}(1+iT)$. Moreover

$$\mathcal{B} \leq (1+tT)^t < (1+(t+1)T)^t + 1 \leq L.$$

Thus the indicated residue is the actual product. An ordinary quotient and remainder, with a strict remainder bound, give its finite polynomial graph. The compiler needs exactly two factorial requests and fourteen binomial requests; all are eliminated by the preceding constructions.

Every addition or multiplication in the resulting fixed circuit can be represented by a natural coordinate z and the quadratic residual $z - a - b$ or $z - ab$. Constants are literal numerals. A finite conjunction of residual equations is replaced by their sum of squares. The inherited exporter first removes 777 nonnegative affine gate coordinates, yielding the 895-coordinate source in Table 1. This exporter is included, and its output is independently regenerated and expanded.

7.1 Certified degree-preserving contraction

Lemma 7.1 (Nonnegative polynomial substitution)**.** *Suppose a current residual is $\varepsilon(x - F(y))$, where $\varepsilon \in \{-1, 1\}$ and F has nonnegative integer coefficients and does not contain x. Delete this residual and substitute F for x in every other residual. The nonnegative zero sets are in bijection, with inverse $x = F(y)$. If F is affine, quadratic residual degree is preserved. The same holds when F is quadratic and x occurs only as a degree-one monomial in every other residual.*

Proof. Any original zero has $x = F(y)$, so projects to a new zero. Conversely, evaluate F on a new natural zero. Its value is natural and restores every original equation. The degree assertions follow by direct substitution. In the quadratic case, the pivot occurs in no product or square elsewhere, so replacement cannot create degree above two. □

This is applied to literal exported residuals, including witness equalities, not only to expressions marked as gates. The algorithm maintains original indices during elimination and records every pivot, its complete restoring polynomial, and its sign. All arithmetic is exact. A support cap chooses which substitutions to attempt; it never truncates a polynomial or discards terms.

With support cap 64, 373 coordinates and residuals are removed from the 895-coordinate source, yielding 522 coordinates and 400 residuals. The caps 16 and 4 yield the other two listed variants. No unused free coordinate is discarded in these three releases, so the recorded projection and restoration are actual bijections of natural zero sets. Restoration is executed in reverse elimination order. Its composed polynomial may have high degree; the final residuals still have degree at most two, which is the degree property relevant to (2).

The independent checker does not invoke the contraction routine or its sparse substitution functions. It reconstructs each pivot from literal coefficient lists, checks nonnegativity, expands each replacement, checks degree after every step, and matches the final renaming and every final residual. Exact squaring then certifies the delivered expanded polynomial.

8 An unrestricted-integer companion without changing degree

To remove ambiguity about the domain, we also deliver an equation over all integers. For each coordinate x_i of P_{522}, introduce three new integer variables a_i, b_i, c_i and define

$$\boxed{P^{\mathbb{Z}}_{2088}(x, a, b, c) = P_{522}(x) + \sum_{i=0}^{521} (8x_i + 1 - a_i^2 - b_i^2 - c_i^2)^2 = 0.} \tag{28}$$

At any integer zero, every guard vanishes, and hence $8x_i + 1 \geq 0$. As x_i is integral, $x_i \geq 0$. Conversely, for every $x_i \geq 0$, the positive integer $8x_i + 1$ is congruent to one modulo eight and is not of the forbidden form $4^a(8b + 7)$. Legendre's three-square theorem therefore supplies a_i, b_i, c_i [11]. This proves the existence equivalence between (28) over $\mathbb{Z}$ and (2) over $\mathbb{N}$.

The integer polynomial is supplied in full, not just as a conversion recipe. Its exact census is

Unrestricted integer coordinates	2,088
Quadratic residuals	922
Expanded monomials	26,791
Exact degree	4
Maximum absolute coefficient	83,901,150

The new monomial a_0^4 has coefficient one, independently certifying exact degree four. The conversion preserves existence; its fibers generally contain several three-square representations, so it is not claimed to be a bijection.

9 Proof of the consistency theorem and its formalization scope

We can now compose the argument without any assumption about whether ZFC is consistent. From a hypothetical natural zero of P_{522}:

1. Restore every contracted coordinate and every removed affine gate. The sum of squares supplies the entire synchronized quadratic system.
2. Apply the common-frame and index theorems to recover every power. Recover every binomial by the common radix, then the factorials and modulus product by their exact graph identities.
3. Apply the shared CRT certificate to obtain every valid bounded local row. The local beta equations and endpoints form one path.
4. Apply the accumulator, carrier, and flow arguments to obtain a finite C_{222} derivation of target code one.
5. Apply the unconditional calibrations (3) to obtain a finite ZFC refutation.

Conversely, a finite ZFC refutation gives a finite C_{222} derivation by the same calibration. Expand it to a tree, choose its label/schema bound, form the carrier and state path, choose a beta encoding, and use the CRT certificate. Evaluate the semantic arithmetic macros, choose their common radix and shared Pell witnesses, and evaluate all gates. Finally project along the recorded contractions. Every choice has been proved to exist, with no bound on its size being assumed.

All these arguments use finite sequences, arithmetic induction, finite CRT, elementary polynomial arithmetic, Pell classification and its finite index criterion, and the stated calculus calibration. They can be carried out in the internal natural numbers of ZFC. In particular, there is no appeal to the arithmetic soundness of ZFC and no restriction of its proof codes to externally standard numerals. This establishes Theorem 1.1 as an ordinary mathematical proof formalizable in ZFC. A standard proof predicate for ZFC may be substituted for the repository's sealed calculus using the usual effective proof translations.

The deliverable does *not* contain a newly checked complete Lean/Rocq derivation of this new polynomial theorem or an emitted object-language ZFC proof term. The immediate inherited Lean statements are included and pinned, but their entire transitive dependency closure has not been rebuilt here. The exact Python audits establish finite identities and artifact correspondence; they do not replace the universal mathematical arguments above. They also do not prove that the equation has no solution.

10 Verification, reproduction, and integration

The release separates universal proofs from finite checks. Its audits cover:

- exact unchanged constructor boundaries and the explicit factored Horn zero-set argument;
- all 388 majorant inequalities, the literal 23-square construction, and positivity of every power exponent;

- exact symbolic recovery of the synchronized power/index equations, the inclusion of all 22 squared ordinates in the common product, and the common binomial radix;
- every contraction pivot, nonnegative restoring polynomial, intermediate degree, final coordinate map, and residual;
- independent expansion and collection of the natural and integer polynomials, exact counts and coefficient heights;
- deterministic regeneration and rejection of deliberately corrupted restoration coefficients, pivot signs, and coordinate maps.

Small Pell, CRT, binomial, and branch fixtures are supplementary regression checks. No finite sample is presented as a proof of the unrestricted consistency equivalence. Both ordinary Python and optimized Python are used; authored validation checks raise explicit exceptions and do not disappear when assertions are disabled.

The archive is self-contained for generation and the finite checks. Python 3.10 or later and its standard library suffice. From its root:

```
python3 verify.py
python3 -O verify.py
python3 code/release.py --output /path/to/fresh-equations
sh build_report.sh
```

The release generator writes a fresh equation tree. The verifier reads the delivered tree, replays certificates, and regenerates in temporary staging; it does not overwrite the mathematical data. The report is compiled from the included LaTeX source with pdfLaTeX.

Integration can preserve this as a new contribution directory with the report, equations, code, proofs, provenance, and receipts together. The main comparison should record the 522-natural quartic and the 2,088-integer quartic separately. The 529- and 596-natural alternatives belong alongside them as expression-size tradeoffs. The source snapshot and the older calibration snapshot are distinct pins and are both recorded.

11 What this iteration contributes

The main change is a general method of organizing certificates across a finite arithmetic circuit. Large bounds are chosen for a whole family; Pell and CRT witnesses are chosen compatibly for that family; and semantic values are separated from auxiliary encoding choices. This eliminates repeated local certificates without weakening their arithmetic content. The small-coefficient, fully expanded direct equation makes the gain concrete: 522 rather than 3,614 natural coordinates, with degree four retained and every coefficient supplied.

The next substantial reduction would have to simplify more of the direct calculus or replace the remaining Pell-index and arithmetic-graph structures. The present contraction order is not proved optimal. The current result already supplies a reusable simultaneous-power theorem, a common-radix binomial theorem, and a complete finite polynomial whose logical meaning is precisely stated. It is a measured improvement over the inspected direct construction, without claiming to resolve the unrestricted search for a globally simplest consistency equation.

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